NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
LAHORE CAMPUS

Linear Algebra Outline according to OBE, FALL-2022

FILE CONTENTS

Outline of Linear Algebra (CS, DS, SE)

DEPARTMENT OF SCIENCES & HUMANITIES

Department	Department of Sciences & Humanities	**Dept. Code**	S & H
Course Title	Linear Algebra	**Course Code**	MT1004
Pre-requisite(s)	-	**Credit Hrs.**	3
Moderator	Dr. Tayyaba/Dr. Akhlaq Ahmad		
Course Instructors	Dr. Akhlaq Ahmad (BCS-3C, 3E), Mr. Saleem Chishti (BSE-3A,3B), Mr. Qasim Noor (BSDS-,1A,1B), Dr. M. Javaid (BCS-3A,3B), Dr. M. Awais Umar (BDS-1C, 1D), Dr. M. Nasir (BCS-3F,3G), Sara Asghar (BCS-,1D).		
Note:	It is a tentative schedule of the course. It may vary (if required).		

Course Objective	The objective is to impart training to the students in this very important branch of Mathematics. Students are expected to learn about system of linear equations, vector spaces, inner products, Eigen values and linear transformations. Attempt will be made to introduce the students to postulation and axiomatic approach in Mathematics. This course also emphasizes the application of linear algebra in science and real life.

No.	Assigned Program Learning Outcome (PLO)	Level	Tool
01			

I = Introduction, R = Reinforcement, E = Evaluation.
A = Assignment, Q = Quiz, M = Midterm, F = Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Tools
01	• Use concept of elementary row operations to find the inverse of square matrices, determinant of a matrix and solving the system of linear equations.	Q1, A1, M1, F
02	• Properties of vectors in 2-space, 3-space and n-space and recognize vector spaces and/or subspaces to compute their bases and its dimension.	Q2, A2, M2, F
03	• Perform Eigen Value analysis and use it to Diagonalize a matrix and/or find its powers.	Q3, A2, M2, F
04	• Identify inner product spaces and/or perform Gram Schmidt process/QR decomposition using inner products.	Q4, M2, A3, F
05	• Express a linear transformation graphically using matrices and to solve problems.	Q3, A3, F

Text Book(s)	**Title**	Elementary Linear Algebra
	Author	Howard A. Anton (Latest Edition)
	Publisher	
Ref. Book(s)	**Title**	Linear Algebra with Applications
	Author	Bernard Kolman (Latest Edition)

Week	Course Contents	Chapter	CLO
01	**System of Linear Equations and Matrices** 1.1 Introduction to linear system 1.2 Gaussian Elimination 1.3 Matrices and Matrix operations	1	01
02	1.4 Inverses; Algebraic properties of Matrices 1.5 Elementary Matrices and Method of finding matrix inverse	1	01
03	1.6 More on linear systems and Invertible Matrices 1.7 Diagonal, Triangular and Symmetric matrices 1.8 Matrix Transformations	1	01, 05
04	**Determinants** 2.1 Determinants by Cofactor Expansion 2.2 Evaluating Determinants by row reduction 2.3 Properties of Determinants; Cramer's rule	2	02
05	**MID TERM-I**		
06	**Euclidean Vector Spaces** 3.1 Vectors in 2-space, 3-space and n-space 3.2 Norm, Dot Product, and Distance in R^n 3.3 Orthogonality of vectors 3.4 The Geometry of Linear Systems 3.5 Cross Product	3	03
07	**General Vector Spaces** 4.1 Real Vector Spaces 4.2 Subspaces	4	05
08	4.3 Linear Independence / Dependence 4.4 Coordinates and Basis for a vector space 4.5 Dimension 4.6 Change of Basis	4	02
09	4.7 Row space, Column Space and Null Space 4.8 Rank and Nullity 4.9 Basic Matrix Transformations in R^2 and R^3 4.10 Properties of Matrix Transformations (if time permits at the end)	4	02
10	**Eigenvalues and Eigen vectors** 5.1 Eigenvalues and Eigenvectors 5.2 Diagonalization of matrices	5	03
11	**Inner Product Spaces** 6.1 Inner product spaces 6.2 Angle and Orthogonality in Inner product Spaces	6	04

12	6.3 Gram-Schmidt Process, QR- Decomposition		
13	**MID TERM II**		
14	**General Linear Transformations** 8.1 General Linear Transformations 8.2 Compositions and Inverse Transformations	8	05
15	8.3 Isomorphism 8.4 Matrices for General Linear Transformations 8.5 Similarity	8	05
If time permits	7.1 Orthogonal Matrices 7.5 Hermitian, Unitary and Normal Matrices		
	FINAL EXAM		

Evaluation Procedure & Marks Distribution:

Assessment Tools	**Total No. of**	**Weightage**
Quizzes	As announced by instructor (3 at least)	10%
Assignments	As announced by **Moderator** (3 at least)	8%
Homework	As announced by instructor (7 at least)	7%
Mid Term Exam	2	25%
Final Exam	1	50%

Grading Scheme: *Relative*